### 9.3.4 Programs using Sets

**Program 1: Program that generate a set of prime numbers and another set of even numbers. Demonstrate the result of union, intersection, difference and symmetirc difference operations.**

**Example**

```
even=set([x*2 for x in range(1,11)])
primes=set()
for i in range(2,20):
        j=2
        f=0
        while j<=i/2:
                if i%j==0:
                        f=1
                j+=1
        if f==0:
                primes.add(i)
print("Even Numbers: ", even)
print("Prime Numbers: ", primes)
print("Union: ", even.union(primes))
print("Intersection: ", even.intersection(primes))
print("Difference: ", even.difference(primes))
print("Symmetric Difference: ", even.symmetric_difference(primes))
```

**Output:**

```
Even Numbers:  {2, 4, 6, 8, 10, 12, 14, 16, 18, 20}
Prime Numbers:  {2, 3, 5, 7, 11, 13, 17, 19}
Union:  {2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20}
Intersection:  {2, 4}
Difference:  {6, 8, 10, 12, 14, 16, 18, 20}
Symmetric Difference:  {3, 5, 6, 7, 8, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20}
```

## 9.4 Dictionaries

### Introduction

In python, a dictionary is a mixed collection of elements. Unlike other collection data types such as a list or tuple, the dictionary type stores a key along with its element. The keys in a Python dictionary is separated by a colon ( : ) while the commas work as a separator for the elements. The key value pairs are enclosed with curly braces { }.